



# Musk ox optimizer (MO): a novel optimization algorithm and its application

Yongliang Yuan<sup>1</sup> · Guangya Chong<sup>1</sup> · Jianji Ren<sup>2</sup> · Wu Zhao<sup>1</sup> · Yanan Li<sup>2</sup> · Zhenxi Wang<sup>3</sup> · Seyedali Mirjalili<sup>4,5</sup>

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## Abstract

This paper proposes a new meta-heuristic algorithm, the Musk Ox Optimizer (MO). The MO mimics the social behavior of musk ox populations, which make migratory, foraging, and defensive choices based on signals sent by “guards” within the population. These behaviors can be translated into mathematical models and eventually an algorithm that effectively address various optimization challenges in different search spaces. To test the MO, 23 benchmark test functions and CEC 2021 are selected. Furthermore, Wilson sign rank tests are used to judge the statistical significance of the results. The findings indicate that the MO outperforms other algorithms in the literature when applied to the two types of test functions. Moreover, the results derived from practical applications also elucidate the advantages that MO holds in resolving problems within unknown search spaces. The source code of MO is publicly available at <https://www.mathworks.com/matlabcentral/fileexchange/177574-musk-ox-optimizer-mo>.

**Keywords** Musk ox optimizer (MO) · Meta-heuristic algorithm · Engineering optimization

✉ Jianji Ren  
renjianji@hpu.edu.cn  
Yongliang Yuan  
yuanyongliang@hpu.edu.cn  
Guangya Chong  
212305020064@home.hpu.edu.cn  
Wu Zhao  
zhaowu@hpu.edu.cn  
Yanan Li  
liyn@hpu.edu.cn  
Zhenxi Wang  
zhenxi23@mails.jlu.edu.cn  
Seyedali Mirjalili  
seyedali.mirjalili@griffithuni.edu.au

<sup>1</sup> School of Mechanical and Power Engineering, Henan Polytechnic University, Jiaozuo 454003, China

<sup>2</sup> School of Software, Henan Polytechnic University, Jiaozuo 454003, China

<sup>3</sup> School of Communication Engineering, Jilin University, Changchun 130012, China

<sup>4</sup> Centre for Artificial Intelligence Research and Optimization, Torrens University Australia, Fortitude Valley, Brisbane, QLD 4006, Australia

<sup>5</sup> University Research and Innovation Center, Obuda University, Budapest 1034, Hungary

## 1 Introduction

Optimization problems can be said to have permeated into various fields. Although traditional optimization techniques are capable of deriving the exact solutions for certain specific problems, they are beset with numerous shortcomings [1]. For example, they tend to become entrapped in local optima and might consume an inordinate amount of time during the problem-solving process. Illustrating this with specific examples, linear programming methods are restricted to handling only linear problems. In the case of gradient-based methods, when confronted with objective functions that possess multiple minima, they are prone to converge to local optima [2].

In view of the shortcomings of traditional optimization algorithms, stochastic optimization algorithms treat optimization problems as black boxes, which to some extent compensates for the deficiencies of traditional optimization algorithms [3]. The advantages of this lie in the fact that there is no need to derive mathematical models when solving problems, and they also possess a high degree of flexibility. Stochastic optimization algorithms handle optimization problems randomly, thereby distinguishing them from traditional optimization techniques [4].

Heuristic and meta-heuristic methods are categorized under the umbrella of stochastic optimization techniques [5]. Due to the numerous limitations of the former, they have not received as much attention as meta-heuristic algorithms. Meta-heuristic algorithms are favored for their straightforwardness, adaptability, and the absence of a requirement for gradient information. They can provide reasonable solutions in a relatively brief timeframe, but cannot guarantee to find the best optimal solution [6–8]. Meta-heuristic strategies can be broadly classified into four main categories, drawing inspiration from various sources [9]:

- (i) **Evolutionary algorithms:** These approaches are inspired by the principles of natural evolution. Among these, GA [10] stands out as a quintessential and highly renowned algorithm, which has witnessed extensive application since its inception. In the natural realm, genetic cross-over and mutation play a pivotal role in sustaining biodiversity and driving biological evolution. GA exploits these two concepts to ensure the algorithm's capacity for exploring the solution space and evading local optima. In addition, the optimizers like DE [11] and ES [12] are also classified as evolutionary methods.
- (ii) **Swarm-based algorithms:** This category of the optimization algorithms principally emulates the social conduct within animal groups, in which every discrete individual exchange information throughout the optimization process. The preeminent one among them is PSO [13]. During the optimization process, individuals search for improved solutions within the search space. They obtain the optimal fitness by using their personal best 'pbest' and the global best 'gbest' solutions. In recent years, researchers have put forward numerous similar optimization methods, such as ABC [14], GWO [15], ASO [16], ChOA [17], and WO [18].
- (iii) **Physics-based algorithms:** GSA [19] and SA [20] belong to this category of algorithms, with their inspirations drawn from the law of universal gravitation and thermodynamics, respectively.
- (iv) **Human-based algorithms:** This category of the algorithms emulates the interactive behaviors within human society. BSO [21] is an instance of such an algorithm, which simulates the process of human cognitive stimulation and innovation. In the algorithmic paradigm, new concepts and solutions are generated via the collective communication and collaboration within the group. It comprises clustering operations and mechanisms for generating new entities, analogous to the group deliberations and the generation of novel viewpoints in a brainstorming gathering. Moreover, ICA [22] and TLBO [23] also fall under the purview of human-based methods.

To address the limitations of the foundational algorithms, researchers have further enhanced their performance by incorporating distinctive strategies. For example, by integrating a learning-imitation strategy into the Alpine Skiing Optimization (ASO), a new optimization algorithm called LISASO has been proposed [24]. To enhance the performance of the Grey Wolf Optimizer algorithm (GWO), researchers integrated an elite opposition-based learning and chaotic k-best gravitational search strategy (EOCS) into GWO, thereby creating the EOCSGWO [25]. Wang proposed an improved Harris Hawks Optimization algorithm (IHHO) that employs spiral search and neighborhood perturbation to enhance the global search capability [26].

Another approach to improving the performance of optimization algorithms is to combine two or more metaheuristic optimizers. This approach combines their strengths and compensates for individual weaknesses, yielding an improved hybrid algorithm. For instance, Seyed Mostafa Bozorgi and Samaneh Yazdani integrated the Whale Optimization Algorithm (WOA) with the Differential Evolution (DE) algorithm and proposed an improved WOA (IWOA), which mitigates WOA's tendency to become trapped in local optima [27]. Shubham Gupta and Kusum Deep combined the Artificial Bee Colony (ABC) algorithm with the Sine Cosine Algorithm (SCA) to create the SCABC [28]. Yin et al. replaced the anisotropic search operator of the Slime Mould Algorithm (SMA) with the search operator from the Equilibrium Optimizer (EO), thereby proposing a novel EOSMA [29].

Although numerous meta-heuristic algorithms have been proposed in the past two decades, they all share a common characteristic: these optimization algorithms involve both exploration and exploitation stages [24, 30–34]. Achieving a balanced strategy between them is a crucial consideration in the examination of meta-heuristic algorithms [35].

In recent years, various optimization problems have exhibited complex characteristics such as high dimensionality, multiple constraints, nonlinearity, and multimodality. To address this challenge, new metaheuristic optimization algorithms have been continuously emerging, as shown in Table 1.

The “No Free Lunch” (NFL) [49] indicates that no single algorithm is universally suitable for all optimization problems. In other words, a certain algorithm may achieve relatively good expected results when dealing with specific optimization problems, but it may perform poorly when confronting other problems. This theory has also fostered advancements in the optimization domain. Researchers are constantly improving existing methods and new algorithms are proposed every year.

This study simulates the migration, perching and defensive behaviors of musk oxen and proposes an innovative

**Table 1** Some of metaheuristic optimization algorithms

| Name | Basic algorithm                       | Year | References |
|------|---------------------------------------|------|------------|
| CSA  | Chameleon Swarm Algorithm             | 2021 | [36]       |
| WHO  | Wild Horse Optimizer                  | 2021 | [37]       |
| LA   | Lichtenberg Algorithm                 | 2021 | [38]       |
| ASO  | Alpine skiing optimization            | 2022 | [16]       |
| HFA  | Human Felicity Algorithm              | 2022 | [39]       |
| HBA  | Honey Badger Algorithm                | 2022 | [40]       |
| CPMA | Coronavirus Mask Protection Algorithm | 2023 | [41]       |
| GLA  | Group Learning Algorithm              | 2023 | [42]       |
| WCO  | Willow Catkin Optimization            | 2023 | [43]       |
| FBO  | Frigatebird optimizer                 | 2024 | [44]       |
| ELO  | Eurasian lynx optimizer               | 2024 | [45]       |
| AMA  | Artificial meerkat algorithm          | 2024 | [46]       |
| FCO  | Fishing cat optimizer                 | 2025 | [47]       |
| DLO  | Draco lizard optimizer                | 2025 | [48]       |

meta-heuristic optimizer, namely, Musk Ox Optimizer (MO). In this work, the “safety signals” and “alert signals” sent by the “guards” are crucial to the MO. When the “alarm signal” exceeds a certain value, they will migrate to a safe area, which is the exploratory phase of the MO. Otherwise, they will enter the roosting phase, which is the exploitation stage of the algorithm. In this stage, the “safety signal” is crucial as it influences the musk oxen’s decision between defensive and foraging activities. The efficiency of the MO is evaluated by 23 benchmark functions, CEC2021 and six traditional structural optimization problems. The findings indicate that the MO demonstrates strong competitiveness when compared to advanced optimization methods.

The main innovative points of MO are as follows:

- The MO can achieve a smooth transition from the exploration phase to the exploitation phase by utilizing the adaptive range changes of the “alert signals”.
- The “alert signals” contribute to the MO’s discovery of promising areas in the search space during the exploration phase.
- The “safety signals” contribute to the MO’s conducting exploitation and local search within the promising search intervals.
- Taking the best approximation of the global optimum as the target point, it is stored in a variable, and this value will not be lost during the optimization process.

The remainder of this study is organized as follows: Sect. 2 describes the inspiration for MO and its mathematical model. Section 3 analyzes the results of MO on several test functions. Section 4 tested the performance of MO in six engineering cases. Section 5 summarizes the content and ideas of the work and presents the future research directions of the MO.

## 2 Musk ox optimizer (MO)

This section first discusses the biological basis of the Musk Ox Optimizer (MO). Subsequently, the mathematical model of the MO is presented.

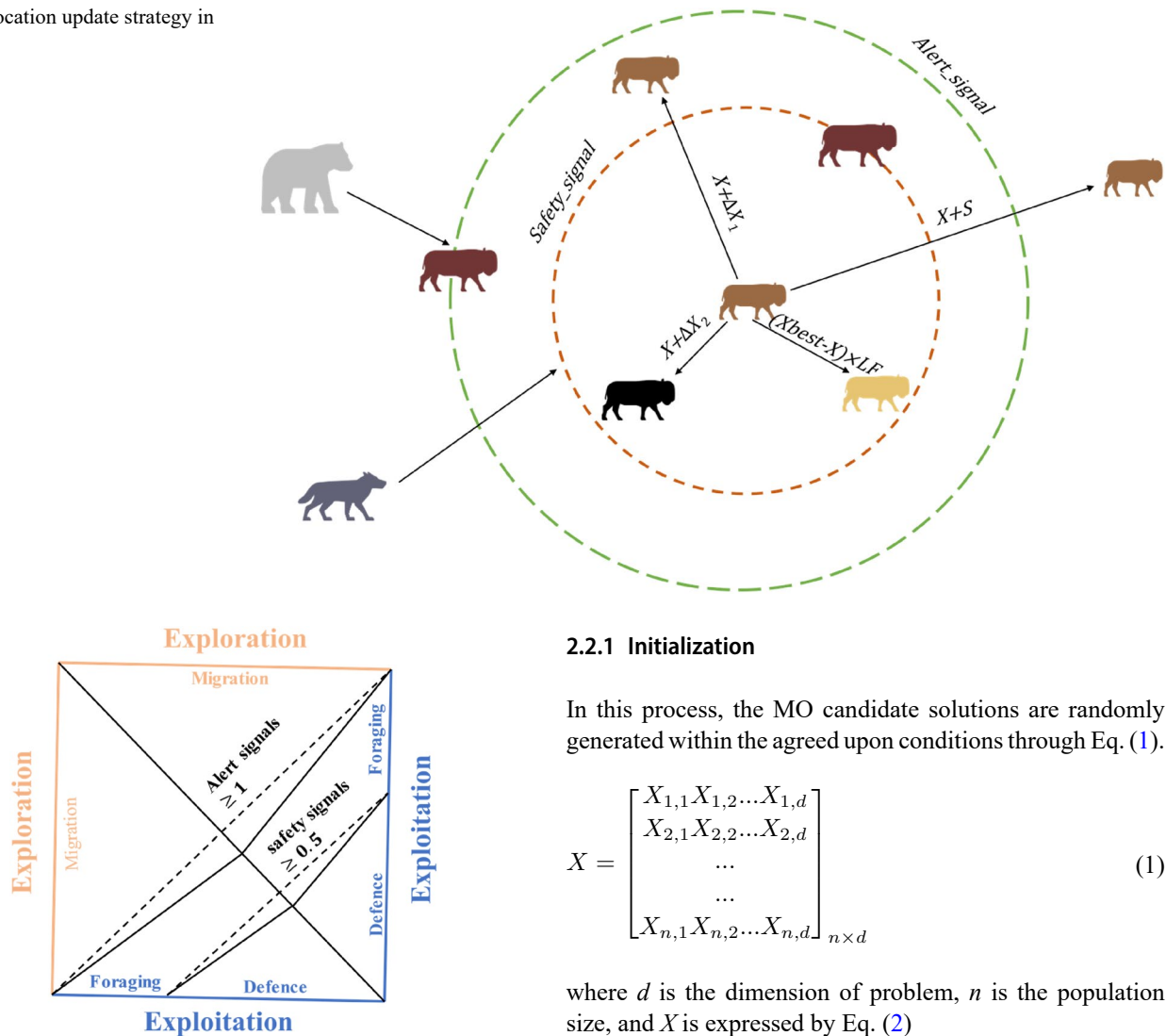
### 2.1 Source of inspiration

The musk ox is a mammal belonging to the genus Musk Ox, family Bovidae, order of hoofed animals [50]. They primarily inhabit the harsh, barren lands of the Arctic tundra regions in northern Canada, Greenland, and Norway. The musk ox is classified as a transitional species between cattle and sheep, despite being commonly referred to as a cow. Like bison, they form herds with a distinct hierarchy, often led by a pregnant female musk ox. This unique leadership structure sets them apart from other herd animals. Musk oxen are brave by nature and do not retreat and run away from danger [51]. When confronted by natural predators, such as arctic wolves, the herd immediately adopts a defensive stance. Adult musk oxen take the forefront, protecting their young in the center, while the bulls on the periphery prepare for surprise attacks with their sharp horns. Their long, thick fur acts as a formidable defense, easily neutralizing enemy attacks. Following the threat, the musk oxen promptly resume their positions, awaiting any further encounters. Their thick and dense downy hair provides ample insulation against the cold and moisture, while the outer layer of coarse long hair serves as protection against rain, snow, and wind. As a result, musk oxen thrive in cold environments but struggle to adapt to warmer climates. Figure 1 presents musk ox.

The MO was used to model the behavior and division of roles in musk ox populations. Specifically, the MO makes assumptions about the social structure and interactions among male, female, and juvenile musk oxen. The algorithm is schematically depicted in Fig. 2, where the red icons representing the “guards” in the musk ox population. The signals from the “guards” have a crucial role for the whole group. The brown and black icons represent musk ox group (All members of the group) and adult male musk oxen, respectively, while the yellow icons represent adult females and immature musk oxen.

### 2.2 Mathematical model

This subsection will conduct mathematical modeling for each stage of the MO. Figure 3 illustrates all phases and corresponding strategies of the MO.

**Fig. 1** Musk ox resting in the snow**Fig. 2** Location update strategy in the MO

### 2.2.1 Initialization

In this process, the MO candidate solutions are randomly generated within the agreed upon conditions through Eq. (1).

$$X = \begin{bmatrix} X_{1,1} & X_{1,2} & \dots & X_{1,d} \\ X_{2,1} & X_{2,2} & \dots & X_{2,d} \\ \dots & \dots & \dots & \dots \\ X_{n,1} & X_{n,2} & \dots & X_{n,d} \end{bmatrix}_{n \times d} \quad (1)$$

where  $d$  is the dimension of problem,  $n$  is the population size, and  $X$  is expressed by Eq. (2)

$$X = MO_{lb} + rand \times (MO_{ub} - MO_{lb}) \quad (2)$$

**Fig. 3** Different phases of MO

where  $MO_{lb}$  is the upper boundary of the problem variable. On the contrary,  $MO_{ub}$  is the lower boundary, rand is a random number between 0 and 1. Additionally,  $r_1, r_2, r_3, r_4, r_5$  and  $r_6$  appearing below are same as rand.

In a musk ox population, the number of adult females and males each make up 40% of the population, while the remaining 20% comprises immature individuals.

### 2.2.2 Alert and safety signals

Musk oxen are very sensitive to danger perception, once danger is detected, the patrollers send out “alert signals” to notify the group to take precautions. In the MO, “alert signals” and “safety signals” are defined as follows:

$$Alert\_signal = A \quad (3)$$

$$\alpha_1 = 1 - t/T \quad (4)$$

$$A = 2 \times \alpha_1 \times (1 - r_1) \quad (5)$$

where  $A$  is alert factor,  $T$  is the maximum iteration, and  $\alpha_1$  gradually drops from 1 to 0 with  $t$ .

The definitions for “safety signals” are as follows:

$$Safety\_signal = r_2 \quad (6)$$

**Table 2** The pseudo-code of the MO

|                                  |                                               |
|----------------------------------|-----------------------------------------------|
| <i>Input: N, T</i>               |                                               |
| 1:                               | <i>Initialize</i>                             |
| 2:                               | <b>While</b> $t \leq T$                       |
| 3:                               | <i>Compute fitness values</i>                 |
| 4:                               | <i>Update the best solution</i>               |
| 5:                               | <b>Exploration:</b>                           |
| 6:                               | <b>If</b> $ Alert\_signal  \geq 1$            |
| 7:                               | <i>Update position by Eq. (8)</i>             |
| 8:                               | <b>End if</b>                                 |
| 9:                               | <b>Exploitation</b>                           |
| 10:                              | <b>If</b> $ Safety\_signal  \geq 0.5$         |
| 11:                              | <i>Update position by Eq. (11)</i>            |
| 12:                              | <b>Else</b>                                   |
| 13:                              | <b>For each male Musk oxen</b>                |
| 14:                              | <i>Update position by Eq. (14)</i>            |
| 15:                              | <b>End For</b>                                |
| 16:                              | <b>For each juvenile and female Musk oxen</b> |
| 17:                              | <i>Update position by Eq. (17)</i>            |
| 18:                              | <b>End For</b>                                |
| 19:                              | <b>End If</b>                                 |
| 20:                              | <b>End While</b>                              |
| <i>Output: the best solution</i> |                                               |

### 2.2.3 Migration (Exploration phase)

The population of musk oxen will migrate to safer places when the threat level becomes too great. The following is a mathematical model of a musk ox during migration.

$$X^{t+1} = X^t + S \quad (7)$$

where,  $x^{t+1}$  denotes the new position and  $x^t$  is the current position of musk oxen,  $S$  is the step length of the musk ox movement as shown in Eq. (8).

$$S = \alpha_2 \wedge 2 \times r_3 \wedge 4 \times (X_a^{-t} - X_b^{-t}) \quad (8)$$

$$\alpha_2 = 1 - \frac{1}{1 + \exp(\frac{T-t}{T} \times 10)} \quad (9)$$

where  $X_a^t$  and  $X_b^t$  are the randomly generated positions of two “guards”,  $\alpha_2$  is the step influence factor.

### 2.2.4 Roosting behavior (Exploitation phase)

When risk factors are not capable of posing a threat to musk oxen, the population will choose to roost. During the roosting phase, musk ox populations also select two different roosting states, foraging and defense, depending on the margin of safety. In this phase, adult males, females, and juvenile will renew their locations and interact with each other.

#### 1) Foraging behavior.

When the safety margin is sufficiently large, musk ox populations will replenish their energy by foraging. At this stage, various individuals will work together to move and forage depending on where other musk oxen are. This helps the musk ox group to find areas that are more suitable for survival.

$$X^{t+1} = X^t + \Delta x_1 \quad (10)$$

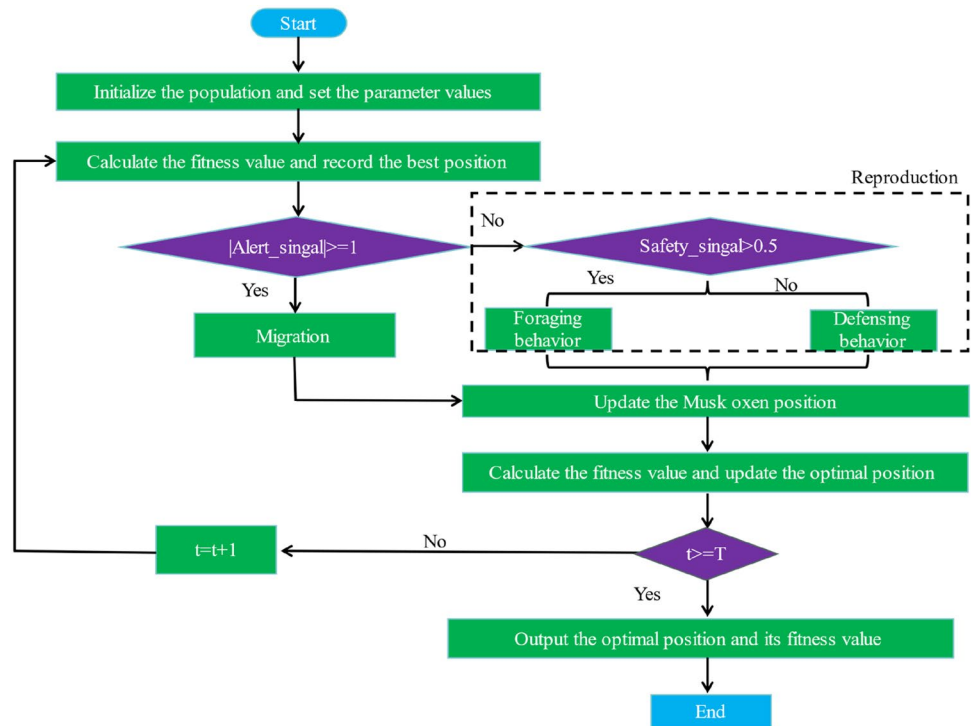
$$\Delta x_1 = rand \times (X_{best}^t - X^t) + r_4 \wedge 3 \times W_i \times (X^t - X_{best}^t) \quad (11)$$

$$W_i = G_1 \times \frac{1}{\exp(\frac{\beta \times t}{T})} \quad (12)$$

where  $X_{best}^t$  is the position of the best musk ox in all iterations,  $W_i$  is the step influence factor,  $G_1, \beta$  are the constants and their values are 0.001 and 0.01, respectively.

#### 2) Defensive behavior.

Musk ox groups inhabit environments where they may encounter natural predators such as polar bears and arctic wolves. In such situations, the herd instinctively gathers together under the guidance of their leader’s powerful roar, forming a united front to defend against the threat. When

**Fig. 4** Flow chart of the MO**Table 3** Parameter values of algorithms

| Algorithm | Parameters                   | Value      | Year |
|-----------|------------------------------|------------|------|
| WOA       | $b$                          | 1          | 2016 |
| PSO       | $w_{max}, w_{min}$           | 0.9, 0.6   | 1995 |
|           | $c_1, c_2$                   | 2, 2       |      |
| BOA       | $c$                          | 0.01       | 2019 |
|           | $a$                          | 0.1        |      |
|           | $p$                          | 0.8        |      |
| MVO       | $WEP_{max}, WEP_{min}$       | 1, 0.2     | 2015 |
|           | $p$                          | 6          |      |
| SSA       | $v_0$                        | 0          | 2020 |
| MFO       | $b$                          | 1          | 2015 |
| DMOA      | $peep$                       | 2          | 2022 |
|           | <i>Number of babysitters</i> | 3          |      |
| GOA       | $f, l$                       | 0.5, 1.5   | 2020 |
|           | $c_{max}, c_{min}$           | 1, 0.00004 |      |
| NGO       | $I$                          | 1 or 2     | 2021 |
| SS        | $v_0$                        | 0          | 2017 |

attacked, the musk oxen utilize their sharp horns as weapons, piercing the bellies of their enemies and throwing them away. When one musk ox falls, the rest quickly fill in the gaps, narrowing the circle, and then continue the fight.

a. Position update of male Musk oxen

$$X^{t+1} = X^t + \Delta x_2 \quad (13)$$

$$\Delta x_2 = r_5 \cdot 2 \times (X_{second}^t - X_{best}^t) + r_5 \times M_i \times (X_{best}^t - X_{worst}^t) \quad (14)$$

$$M_i = G_2 \times \frac{1}{\exp(\frac{\beta \times t}{T})} \quad (15)$$

where  $X_{second}^t$  is the position of the second Musk oxen in all iterations,  $X_{worst}^t$  is the position of the worst musk oxen in all iterations,  $M_i$  is the step control factor,  $G_2$  is a constant and its value is 100.

b. Positions update of female and juvenile Musk oxen.

$$X^{t+1} = r_6 \times (X_{best}^t - X^t) \times LF \quad (16)$$

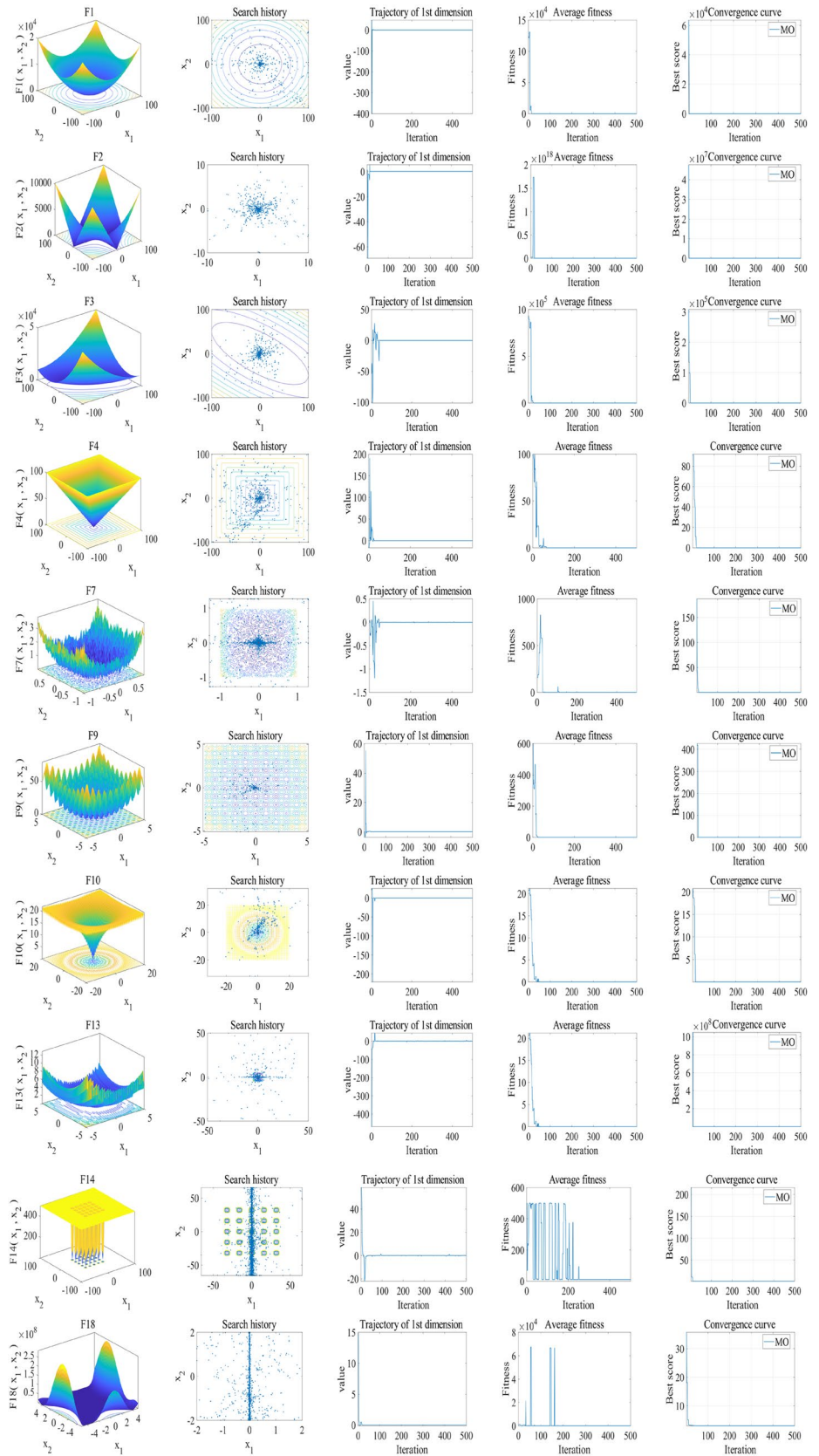
where  $LF$  is a random walk (Levy flight) function.

$$Levy(a) = 0.05 \times \frac{x}{|y|^{\frac{1}{a}}} \quad (17)$$

where  $x$  and  $y$  represent  $x \sim N(0, \sigma_x^2)$  and  $y \sim N(0, \sigma_y^2)$ , respectively.  $a$  is a constant ( $a=1.5$ ).

$$\sigma_x = \left[ \frac{\Gamma(1+\alpha) \sin(\frac{\pi\alpha}{2})}{\Gamma(\frac{1+\alpha}{2}) \alpha \times 2^{\frac{(\alpha-1)}{2}}} \right]^{\frac{1}{\alpha}}, \sigma_y = 1 \quad (18)$$

where  $\sigma_x^2$  and  $\sigma_y^2$  are the standard deviations,  $\Gamma(x) = (x+1)!$ .

**Fig. 5** Qualitative results of the MO

**Table 4** Results of F1-F7

| Function |     | WOA      | GWO             | PSO      | BOA      | MFO      | MVO      | SSA             | MO              |
|----------|-----|----------|-----------------|----------|----------|----------|----------|-----------------|-----------------|
| F1       | Avg | 1.92E-98 | 2.96E-41        | 1.29E-05 | 1.07E-11 | 2.50E+03 | 2.10E+00 | 5.09E-159       | <b>0.00E+00</b> |
|          | Std | 1.45E-98 | 3.79E-41        | 1.25E-05 | 6.54E-13 | 4.52E+03 | 8.99E-01 | 8.64E-159       | <b>0.00E+00</b> |
| F2       | Avg | 9.54E-58 | 4.74E-24        | 5.97E-04 | 3.46E-09 | 4.42E+02 | 4.32E+00 | 2.08E-73        | <b>0.00E+00</b> |
|          | Std | 1.85E-57 | 3.29E-24        | 5.79E-04 | 1.17E-09 | 2.77E+02 | 1.77E+01 | 4.54E-74        | <b>0.00E+00</b> |
| F3       | Avg | 4.67E+04 | 3.00E-11        | 6.31E+02 | 1.09E-11 | 1.28E+04 | 2.89E-01 | 4.39E-116       | <b>0.00E+00</b> |
|          | Std | 5.00E+04 | 7.46E-11        | 3.16E+02 | 8.37E-13 | 8.94E+03 | 1.14E-01 | 8.17E-116       | <b>0.00E+00</b> |
| F4       | Avg | 4.13E+01 | 1.51E-10        | 3.69E+00 | 5.19E-09 | 6.19E+01 | 7.17E-01 | 1.25E-76        | <b>0.00E+00</b> |
|          | Std | 5.86E+01 | 2.04E-10        | 1.33E+00 | 3.71E-10 | 7.00E+01 | 3.03E-01 | 4.22E-75        | <b>0.00E+00</b> |
| F5       | Avg | 2.68E+01 | <b>2.63E+01</b> | 3.25E+02 | 2.89E+01 | 8.50E+03 | 3.00E+02 | <b>5.98E-05</b> | 2.85E+01        |
|          | Std | 1.55E-01 | 7.90E-01        | 8.73E+02 | 2.35E-02 | 2.57E+04 | 5.61E+02 | <b>4.86E-05</b> | 2.57E-01        |
| F6       | Avg | 4.71E-03 | 3.31E-01        | 6.20E-06 | 5.12E+00 | 1.01E+03 | 3.82E-01 | <b>6.45E-11</b> | 2.65E+00        |
|          | Std | 2.40E-03 | 3.72E-01        | 9.99E-06 | 5.49E-01 | 3.11E+03 | 9.06E-02 | <b>1.59E-10</b> | 1.35E-01        |
| F7       | Avg | 6.73E-04 | 6.44E-04        | 2.73E-02 | 9.30E-04 | 1.83E+00 | 1.34E-02 | 1.79E-04        | <b>5.75E-06</b> |
|          | Std | 6.70E-04 | 3.41E-04        | 9.54E-03 | 2.06E-04 | 3.62E+00 | 5.48E-03 | 1.49E-04        | <b>4.88E-06</b> |

**Table 5** Results of F8-F13

| Function |     | WOA              | GWO       | PSO             | BOA       | MFO       | MVO       | SSA       | MO              |
|----------|-----|------------------|-----------|-----------------|-----------|-----------|-----------|-----------|-----------------|
| F8       | Avg | <b>-1.26E+04</b> | -6.68E+03 | -8.99E+03       | -4.17E+03 | -8.48E+03 | -7.97E+03 | -8.63E+03 | -8.36E+03       |
|          | Std | 1.00E+03         | 6.39E+02  | 6.33E+02        | 2.93E+02  | 8.68E+02  | 7.70E+02  | 7.17E+02  | <b>2.06E+02</b> |
| F9       | Avg | 2.84E-15         | 1.02E+00  | 4.14E+02        | 2.60E-12  | 1.33E+02  | 9.65E+02  | 4.26E-01  | <b>0.00E+00</b> |
|          | Std | 1.27E-14         | 2.81E+00  | 8.71E+01        | 1.02E-11  | 2.98E+01  | 2.47E+02  | 7.36E-01  | <b>0.00E+00</b> |
| F10      | Avg | 4.71E-15         | 2.76E-14  | 1.54E+00        | 4.96E-09  | 5.41E+00  | 1.06E+00  | 1.30E-01  | <b>4.44E-16</b> |
|          | Std | 2.18E-15         | 3.11E-15  | 3.31E+00        | 4.20E-10  | 8.27E+00  | 7.61E-01  | 9.00E-01  | <b>0.00E+00</b> |
| F11      | Avg | 4.32E-03         | 3.50E-03  | 1.25E-02        | 2.79E-12  | 9.73E+00  | 6.39E-01  | 9.35E-03  | <b>0.00E+00</b> |
|          | Std | 1.93E-02         | 2.79E-03  | 1.23E-02        | 1.06E-12  | 2.77E+01  | 1.73E-02  | 4.32E-03  | <b>0.00E+00</b> |
| F12      | Avg | <b>7.53E-04</b>  | 2.04E-02  | 1.03E-02        | 3.97E-01  | 2.43E+00  | 5.57E-01  | 9.56E+00  | 5.81E-02        |
|          | Std | <b>1.05E-03</b>  | 1.41E-02  | 3.19E-02        | 1.01E-01  | 1.56E+00  | 6.22E-01  | 1.75E+00  | 4.88E-02        |
| F13      | Avg | 2.98E-01         | 1.84E-01  | <b>1.13E-03</b> | 2.41E-01  | 3.02E+00  | 5.47E-02  | 5.36E-03  | 2.49E+00        |
|          | Std | 3.17E-01         | 1.61E-01  | <b>3.39E-03</b> | 3.64E-01  | 2.35E+00  | 2.07E-02  | 7.64E-03  | 1.08E-02        |

### 2.3 The procedure of MO

In the MO, the alert signal and the safety signal are of crucial importance as they influence various stages of the MO. Musk oxen will exhibit corresponding social behaviors according to the magnitudes of the two signals. Table 2; Fig. 4 include detailed descriptions of the MO's pseudo-code and flowchart, respectively.

### 2.4 Computational complexity

The computational complexity of the MO is mainly affected by three factors: (i) initialization, (ii) fitness evaluation, and (iii) musk ox position update. Suppose the population comprises  $N$  musk oxen during optimization; the initialization phase's complexity is  $O(N)$ , and the update mechanism's is  $O(N \times T) + O(N \times T \times \text{dim})$ , with 'dim' being the problem's dimension and 'T' the maximum iterations. Consequently, the MO's complexity can be expressed as  $O(N \times (T + T \times \text{dim} + 1))$ .

The subsequent text employs various benchmark functions and four practical scenarios to assess, scrutinize, and validate the applicability of the MO.

## 3 Experimental outcomes and discussions

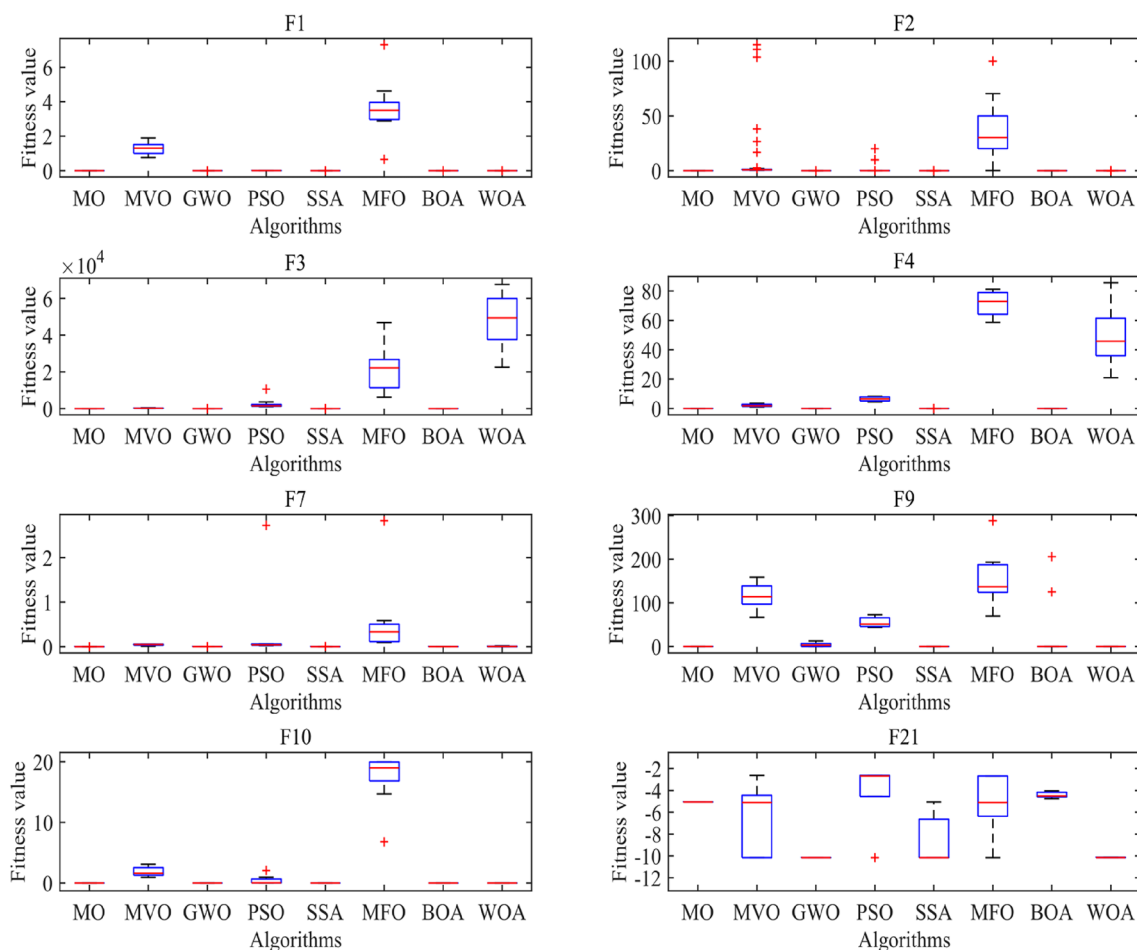
In this section, 23 benchmark functions and CEC 2021 will be utilized to verify the performance of MO. Specifically, in this work, MO and other renowned optimization algorithms are experimented with separately, and the results are thoroughly compared. To ensure a fair comparison among these optimization algorithms, the maximum number of evaluations  $\text{Max\_evaluations} = 50,000$ , and each optimization algorithm is run independently 30 times. Table 3 displays the parameters for each algorithm.

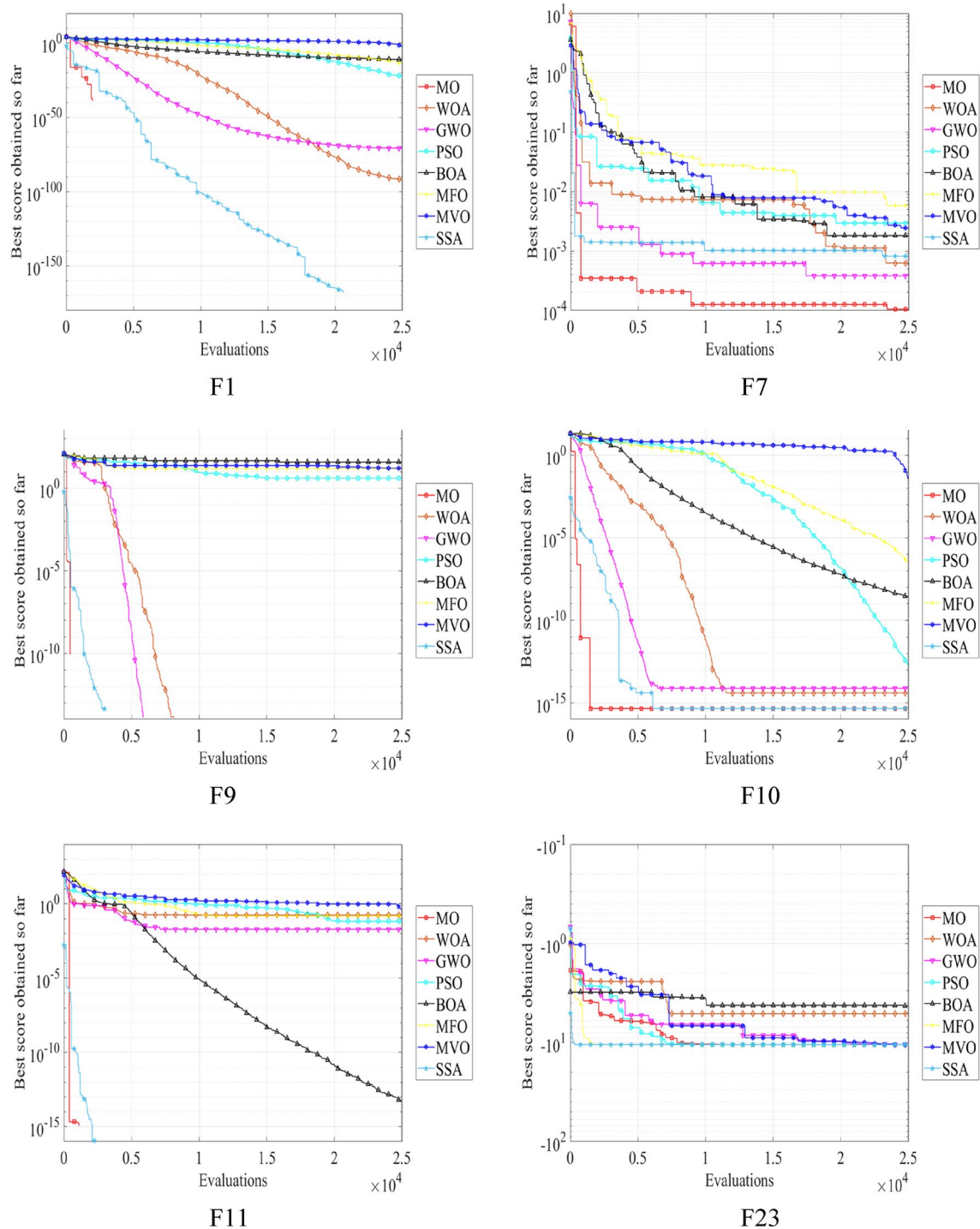
### 3.1 23 benchmark functions analysis

These functions can be classified into single-peak (F1-F7), multi-peak (F8-F13), and low-dimensional multi-peak (F14-F23) benchmark functions based on their shapes,

**Table 6** Results of F14-F23

| Function |     | WOA              | GWO              | PSO              | BOA             | MFO              | MVO              | SSA              | MO               |
|----------|-----|------------------|------------------|------------------|-----------------|------------------|------------------|------------------|------------------|
| F14      | Avg | 1.09E-01         | 2.18E+00         | <b>9.98E-01</b>  | <b>9.98E-01</b> | <b>9.98E-01</b>  | <b>9.98E-01</b>  | 3.63E+00         | <b>9.98E-01</b>  |
|          | Std | 4.43E-01         | 9.97E-01         | <b>7.40E-11</b>  | 2.43E-03        | 4.19E-01         | 1.53E-11         | 4.69E+00         | 3.12 E-01        |
| F15      | Avg | 5.13E-04         | <b>3.08E-04</b>  | 4.08E-04         | 3.44E-04        | 7.38E-04         | 3.60E-03         | <b>3.08E-04</b>  | <b>3.08E-04</b>  |
|          | Std | 5.05E-04         | <b>9.13E-09</b>  | 2.08E-04         | 2.03E-05        | 3.63E-04         | 8.25E-03         | 1.31E-04         | 4.27E-05         |
| F16      | Avg | <b>-1.03E+00</b> | <b>-1.03E+00</b> | <b>-1.03E+00</b> | -7.12E-01       | <b>-1.03E+00</b> | <b>-1.03E+00</b> | <b>-1.03E+00</b> | <b>-1.03E+00</b> |
|          | Std | 1.73E-11         | 6.83E-09         | 2.28E-16         | 2.13E-02        | 2.28E-16         | 4.80E-07         | 2.43E-16         | <b>1.05E-16</b>  |
| F17      | Avg | <b>3.98E-01</b>  | <b>3.98E-01</b>  | <b>3.98E-01</b>  | 6.76E+01        | <b>3.98E-01</b>  | <b>3.98E-01</b>  | <b>3.98E-01</b>  | <b>3.98E-01</b>  |
|          | Std | 4.49E-07         | 1.80E-03         | <b>0.00E+00</b>  | 9.45E+01        | <b>0.00E+00</b>  | 8.99E-08         | <b>0.00E+00</b>  | <b>0.00E+00</b>  |
| F18      | Avg | <b>3.00E+00</b>  | <b>3.00E+00</b>  | <b>3.00E+00</b>  | <b>3.01E+00</b> | <b>3.00E+00</b>  | <b>3.00E+00</b>  | <b>3.00E+00</b>  | <b>3.00E+00</b>  |
|          | Std | 8.36E-07         | 2.80E-06         | 2.88E-16         | 1.04E-02        | 4.25E-15         | 2.31E-07         | 2.46E-15         | <b>1.86E-16</b>  |
| F19      | Avg | <b>-3.86E+00</b> | <b>-3.86E+00</b> | <b>-3.86E+00</b> | -6.44E+01       | <b>-3.86E+00</b> | <b>-3.86E+00</b> | <b>-3.86E+00</b> | <b>-3.86E+00</b> |
|          | Std | 1.74E-03         | 2.78E-03         | <b>2.28E-15</b>  | 5.08E-01        | 6.25E-15         | 2.12E-07         | 7.36E-15         | 7.40E-06         |
| F20      | Avg | -3.27E+00        | -3.25E+00        | -3.27E+00        | -9.51E-01       | <b>-3.22E+00</b> | <b>-3.32E+00</b> | -3.27E+00        | <b>-3.32E+00</b> |
|          | Std | 7.35E-02         | 7.34E-01         | 6.06E-02         | 3.12E-01        | 5.35E-02         | 5.63E-02         | 6.02E-02         | <b>5.01E-03</b>  |
| F21      | Avg | -9.12E+00        | -9.89E+00        | -7.63E+00        | -6.54E+00       | -8.28E+00        | -7.86E+00        | -1.02E+01        | <b>-1.02E+01</b> |
|          | Std | 2.08E+00         | 1.14E+00         | 2.95E+00         | <b>5.53E-01</b> | 4.12E+00         | 3.62E+00         | 2.07E+00         | 2.73E+00         |
| F22      | Avg | -9.22E+00        | -1.04E+01        | -8.84E+00        | -6.64E+00       | -9.20E+00        | -3.43E+00        | -1.04E+01        | <b>-1.04E+01</b> |
|          | Std | 2.46E+00         | <b>3.78E-04</b>  | 2.83E+00         | 9.49E-01        | 4.56E+00         | 1.09E+00         | 1.46E+00         | 2.57E+00         |
| F23      | Avg | -8.83E+00        | -1.03E+01        | -9.73E+00        | -7.24E+00       | -1.02E+01        | -7.06E+00        | <b>-1.05E+01</b> | <b>-1.05E+01</b> |
|          | Std | 3.10E+00         | 1.20E+00         | 1.96E+00         | <b>5.74E-01</b> | 2.05E+00         | 2.26E+00         | 2.38E+00         | 1.18E+00         |

**Fig. 6** Boxplot of benchmark functions

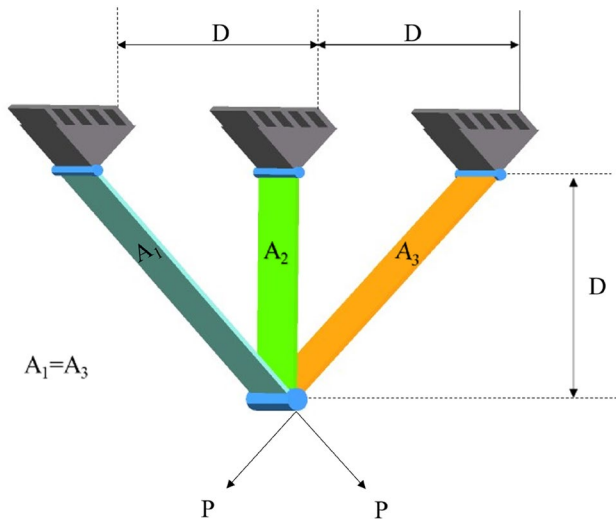


**Fig. 7** The convergence curves of various algorithms

characteristics, complexity, and dimensionality. Here, WOA [52], GWO [15], PSO [13], BOA [53], MFO [54], GA [10], MVO [55] and SSA [56] will be executed alongside MO. We utilize the outcomes of the executions of these approaches for comparison with those of MO, enabling a more intuitive perception of MO's performance.

### 3.1.1 Qualitative analysis

The qualitative solution results include search history, search trajectory, average fitness and convergence curve. Figure 5 presents the qualitative results of MO on 10 test functions.



**Fig. 8** A model of the TTD

The search history graph displays the position history of musk oxen. For unimodal functions F1, F2, F3, F4, and F7, the historical position benchmarks of the musk ox population are distributed around the theoretical optimal value of 0. For multimodal functions F9, F10, F13, F14, and F18, the sampling points are mainly concentrated near the theoretical optimal solutions and local extreme values. This proves that the MO tends to converge to favorable positions throughout the iterative process.

The trajectory curves can illustrate the search processes at different stages. In the early stage of iteration, the trajectory curve of MO always shows obvious fluctuations. This phenomenon indicates that the musk ox population conducts a large-scale search in the search space, possessing excellent global exploration ability, which helps prevent the MO from falling into local optima in the later iterations. In the later stage of iteration, the trajectory curve of MO gradually tends to stabilize, which proves the strong convergence ability of the MO.

The average fitness curve reflects the changes in the average fitness of search agents throughout the iterative process. In the early stage of iteration, the value of the MO is tremendous, which indicates that the algorithm has strong search ability. Subsequently, the average fitness curve of the MO drops rapidly, indicating that the search agents have converged near the theoretical optimal solution.

The convergence curve of the MO always shows a downward trend and completes this process in a short time. This indicates that the MO can quickly find the optimal solution.

### 3.1.2 Statistical results

Examining Tables 4 and 5, and 6 reveals that the MO performs better in 31 out of 46 indicators and shows outstanding performance on 19 out of 23 functions.

Among the 14 indicators of the F1-F7 test functions, MO achieved the best values in 10 indicators. Specifically, MO performed optimally in all indicators of the F1, F2, F3, F4 and F7 test functions, showing significant advantages. SSA showed the best experimental results in F5 and F6, but its experimental results in the remaining test functions were inferior to those of MO, ranking second among the 8 optimization algorithms. PSO, MFO and MVO performed poorly overall.

In the experimental results of F8-F13, MO obtained the first place in all indexes of F9, F10 and F11. The WOA performed better on the F8 and F12 test functions, while the PSO obtained the best values in two of the indicators of the F13 test function.

In the 20 indicators of the experimental results of the F14 - F23 test functions, MO achieved the best values in 14 of them. Specifically, MO outperformed other algorithms in all 10 average value indicators, and it also held an advantage over other approaches in 4 of the 10-variance metrics.

In general, the MO has absolute advantages over other optimization algorithms in solving both unimodal and multimodal functions. This is attributed to the migration strategy and roosting behavior of the MO, which enable a relatively balanced exploration and exploitation phase of the algorithm.

### 3.1.3 Boxplot analysis

The boxplots in Fig. 6 display the various methods employed on 8 test functions. It is evident that MFO exhibits discrete characteristics on the eight presented test functions. All the other optimization algorithms except MO exhibit discrete phenomena in these eight test functions, and some are even accompanied by the generation of singular values (red “+”). These findings suggest that the MO is capable of sustaining a high degree of stability even when confronted with complex challenges.

### 3.1.4 Convergence analysis

Although the above analysis can show to a certain extent that the MO is superior to the other seven optimization algorithms, it is necessary to carry out convergence analysis to better reflect the search performance of the MO. The convergence curve can intuitively present the dynamic process of iterative optimization of various algorithms on test functions. Figure 8 shows the convergence curves of MO, WOA,

**Table 7** Results of the Wilson sign rank tests

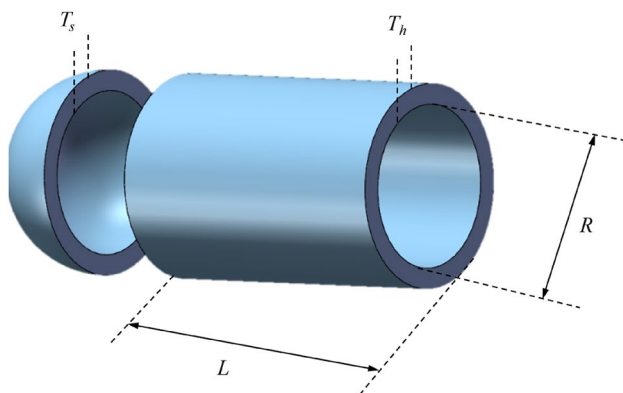
| Function | WOA-MO    |   | GWO-MO    |   | PSO-MO    |   | BOA-MO    |   | MFO-MO    |   | MVO-MO    |   | SSA-MO    |   |
|----------|-----------|---|-----------|---|-----------|---|-----------|---|-----------|---|-----------|---|-----------|---|
|          | P         | H | P         | H | P         | H | P         | H | P         | H | P         | H | P         | H |
| F1       | 3.99E-159 | + | 5.22E-159 | + | 7.59E-174 | + | 1.48E-161 | + | 8.99E-166 | + | 3.07E-172 | + | 4.25E-81  | + |
| F2       | 1.09E-140 | + | 9.34E-162 | + | 1.08E-167 | + | 3.05E-163 | + | 1.64E-164 | + | 3.06E-151 | + | 2.56E-107 | + |
| F3       | 7.88E-177 | + | 1.23E-151 | + | 9.81E-153 | + | 3.23E-30  | + | 5.64E-169 | + | 1.39E-164 | + | 2.52E-99  | + |
| F4       | 3.20E-173 | + | 2.33E-158 | + | 6.12E-169 | + | 1.87E-159 | + | 7.59E-178 | + | 5.07E-171 | + | 1.92E-124 | + |
| F5       | 5.05E-77  | + | 5.12E-107 | + | 2.75E-153 | + | 6.93E-130 | + | 4.84E-152 | + | 2.12E-151 | + | 3.45E-164 | + |
| F6       | 1.35E-131 | + | 1.32E-126 | + | 3.37E-70  | + | 3.68E-65  | + | 5.28E-105 | + | 6.71E-126 | + | 1.23E-169 | + |
| F7       | 2.05E-140 | + | 7.57E-138 | + | 4.79E-162 | + | 1.02E-143 | + | 7.27E-160 | + | 3.84E-158 | + | 6.23E-89  | + |
| F8       | 1.68E-162 | + | 2.03E-27  | + | 5.70E-42  | + | 2.73E-51  | + | 5.14E-120 | + | 7.10E-24  | + | 4.36E-151 | + |
| F9       | 8.38E-158 | + | 7.13E-168 | + | 1.64E-180 | + | 1.04E-168 | + | 2.80E-180 | + | 1.10E-180 | + | 1.06E-01  | + |
| F10      | 5.41E-160 | + | 4.48E-159 | + | 1.45E-160 | + | 5.22E-163 | + | 2.57E-178 | + | 1.40E-175 | + | 3.12E-01  | - |
| F11      | 3.93E-162 | + | 2.95E-162 | + | 2.61E-165 | + | 2.34E-163 | + | 8.29E-173 | + | 3.71E-171 | + | 5.12E-01  | - |
| F12      | 4.23E-143 | + | 4.51E-145 | + | 5.67E-06  | + | 3.68E-56  | + | 1.94E-156 | + | 1.47E-153 | + | 6.32E-165 | + |
| F13      | 3.27E-139 | + | 6.88E-141 | + | 1.10E-06  | + | 3.37E-33  | + | 7.22E-154 | + | 1.24E-72  | + | 2.56E-165 | + |
| F14      | 2.95E-148 | + | 1.84E-142 | + | 3.16E-158 | + | 1.44E-151 | + | 3.99E-141 | + | 4.08E-154 | + | 1.32E-81  | + |
| F15      | 6.28E-152 | + | 1.09E-117 | + | 3.04E-74  | + | 1.44E-94  | + | 8.31E-01  | + | 3.37E-138 | + | 1.21E-158 | + |
| F16      | 3.55E-24  | + | 1.91E-62  | + | 5.34E-17  | + | 2.23E-48  | + | 1.70E-93  | + | 3.81E-76  | + | 2.01E-107 | + |
| F17      | 5.36E-133 | + | 9.57E-129 | + | 1.17E-138 | + | N/A       | = | 4.91E-31  | + | 1.62E-39  | + | 4.56E-32  | + |
| F18      | 3.64E-133 | + | 3.99E-74  | + | 1.10E-14  | + | 2.86E-85  | + | 6.00E-134 | + | 7.53E-07  | + | 1.44E-149 | + |
| F19      | 4.72E-16  | + | 7.70E-27  | + | 1.67E-64  | + | 1.12E-06  | + | 2.36E-150 | + | 1.53E-07  | + | 2.85E-161 | + |
| F20      | 9.60E-155 | + | 5.97E-156 | + | 2.18E-17  | + | 8.99E-10  | + | 5.44E-129 | + | 7.95E-47  | + | 4.59E-158 | + |
| F21      | 3.95E-158 | + | 2.11E-101 | + | 9.89E-95  | + | 2.49E-89  | + | 2.06E-130 | + | 1.66E-10  | + | 3.12E-163 | + |
| F22      | 1.08E-152 | + | 3.02E-121 | + | 1.21E-101 | + | 9.98E-119 | + | 1.14E-133 | + | 1.34E-25  | + | 6.02E-165 | + |
| F23      | 1.75E-156 | + | 6.27E-129 | + | 3.88E-118 | + | 8.85E-64  | + | 7.00E-133 | + | 2.50E-17  | + | 2.56E-164 | + |

**Table 8** Results for CEC 2021

| Function |     | DMOA     | GOA      | MFO      | MVO      | NGO             | SA       | SO        | SS       | AO              | MO               |
|----------|-----|----------|----------|----------|----------|-----------------|----------|-----------|----------|-----------------|------------------|
| CF1      | Avg | 2.12E-17 | 4.63E+03 | 3.46E+03 | 6.78E+04 | 3.96E-96        | 1.51E-08 | 3.73E-118 | 6.84E+02 | 6.42E-102       | <b>1.02E-236</b> |
|          | Std | 3.02E-18 | 2.60E+03 | 4.80E+02 | 3.71E+03 | 3.06E-96        | 1.36E-07 | 1.55E-117 | 3.26E+03 | 9.08E-102       | <b>2.36E-297</b> |
| CF2      | Avg | 2.11E+03 | 8.04E+03 | 7.02E+02 | 1.08E+03 | 7.23E-11        | 5.26E+00 | 4.87E+00  | 2.03E+03 | <b>0.00E+00</b> | <b>0.00E+00</b>  |
|          | Std | 3.46E+02 | 1.31E+01 | 7.89E+01 | 7.14E+01 | 1.56E-12        | 8.96E+00 | 5.49E+01  | 5.12E+01 | <b>0.00E+00</b> | <b>0.00E+00</b>  |
| CF3      | Avg | 4.63E+00 | 7.81E+00 | 8.63E+01 | 4.31E+02 | <b>0.00E+00</b> | 2.78E+00 | 3.12E+01  | 3.05E+00 | <b>0.00E+00</b> | <b>0.00E+00</b>  |
|          | Std | 6.32E+00 | 1.15E+00 | 6.86E+02 | 6.93E+01 | <b>0.00E+00</b> | 3.13E+01 | 7.35E+00  | 1.56E+01 | <b>0.00E+00</b> | <b>0.00E+00</b>  |
| CF4      | Avg | 7.56E-01 | 1.60E+01 | 1.15E+00 | 1.35E+00 | 1.96E-03        | 6.63E-01 | 3.50E-03  | 3.47E+00 | <b>0.00E+00</b> | <b>0.00E+00</b>  |
|          | Std | 5.04E-01 | 9.45E-01 | 1.23E-01 | 6.45E-01 | 1.56E-02        | 6.13E-01 | 2.04E-02  | 1.32E+00 | <b>0.00E+00</b> | <b>0.00E+00</b>  |
| CF5      | Avg | 3.22E+00 | 3.03E+03 | 3.59E+02 | 7.02E+02 | 9.47E-20        | 3.99E+00 | 5.14E+01  | 2.12E+03 | 1.28E+01        | <b>0.00E+00</b>  |
|          | Std | 7.55E+00 | 9.78E+03 | 5.02E+02 | 6.12E+02 | 3.59E-20        | 5.75E+00 | 7.79E+01  | 6.45E+03 | 1.81E+01        | <b>0.00E+00</b>  |
| CF6      | Avg | 3.56E-01 | 1.36E+01 | 6.17E+01 | 2.65E+01 | 1.52E-02        | 3.57E+00 | 8.30E-01  | 1.23E+01 | 2.67E-05        | <b>0.00E+00</b>  |
|          | Std | 1.32E-01 | 4.02E+01 | 9.73E+01 | 4.63E+01 | 6.93E-02        | 5.51E+00 | 2.45E+01  | 3.89E+01 | 3.77E-05        | <b>0.00E+00</b>  |
| CF7      | Avg | 3.56E-01 | 1.03E+04 | 3.67E+01 | 1.63E+01 | 3.85E-04        | 6.03E+01 | 5.90E+00  | 7.57E+03 | 1.13E-62        | <b>0.00E+00</b>  |
|          | Std | 3.89E-02 | 2.89E+03 | 4.32E+01 | 6.85E+01 | 1.23E-04        | 9.63E+00 | 1.96E+01  | 2.88E+03 | 13.8E-62        | <b>0.00E+00</b>  |
| CF8      | Avg | 4.87E+02 | 7.56E+01 | 3.89E+02 | 9.56E+02 | <b>0.00E+00</b> | 3.56E+01 | 3.16E+00  | 8.56E+02 | <b>0.00E+00</b> | 2.34E-137        |
|          | Std | 6.36E+01 | 3.79E+01 | 4.64E+02 | 5.12E+02 | <b>0.00E+00</b> | 4.57E+02 | 8.74E+00  | 3.07E+02 | <b>0.00E+00</b> | 5.12E-156        |
| CF9      | Avg | 6.56E+01 | 7.63E+00 | 9.12E-01 | 9.32E+01 | 9.12E-15        | 3.97E+02 | 4.85E-116 | 2.02E+00 | 2.49E-156       | <b>0.00E+00</b>  |
|          | Std | 8.79E+01 | 6.52E+00 | 1.69E-01 | 2.12E+00 | 8.02E-16        | 3.67E+02 | 3.61E-115 | 5.19E+00 | 3.26E-156       | <b>0.00E+00</b>  |
| CF10     | Avg | 6.10E+00 | 3.48E+00 | 3.94E+01 | 3.56E+01 | 9.06E+01        | 1.20E+01 | 9.43E+01  | 3.08E+02 | 1.23E-127       | <b>0.00E+00</b>  |
|          | Std | 7.01E+01 | 7.33E-01 | 3.18E+00 | 7.02E+00 | 3.37E+01        | 5.60E+01 | 1.62E+01  | 2.03E+00 | 1.75E-127       | <b>0.00E+00</b>  |

**Table 9** Results of the TTD

| Algorithm   | $x_1$       | $x_2$             | $F(x)^*$    |
|-------------|-------------|-------------------|-------------|
| DEDS [64]   | 0.78867513  | 0.40824828        | 263.89584   |
| SSA [56]    | 0.78866541  | 0.408275784       | 263.89584   |
| MBA [65]    | 0.7885650   | 0.4085597         | 263.89585   |
| PSO-DE [66] | 0.7886751   | 0.4082482         | 263.89584   |
| HHO [67]    | 0.788662816 | 0.408283133832900 | 263.89584   |
| MFO [54]    | 0.788244771 | 0.409466905784741 | 263.8959797 |
| MO          | 0.78877016  | 0.40797959        | 263.89584   |

**Fig. 9** A model of the PVD

GWO, PSO, BOA, MFO, MVO and SSA on F1, F7, F9, F10, F11 and F23. Through careful observation and analysis of Fig. 8, the following conclusions can be drawn:

a): Compared with the other 7 optimization algorithms, the MO has the fastest convergence speed in all test functions except F23.

b): MVO, BOA, PSO and MFO exhibit the worst convergence performance on these test functions. The main reason is that the exploration and exploitation stages fail to achieve a reasonable balance throughout the iterative process. In contrast, GWO, WOA and SSA show better convergence accuracy and stability.

c): In the experimental results on the F1, F7, F9, F10, and F11 test functions, the MO demonstrates the optimal accuracy and convergence speed. Regarding the convergence results of F23, among all algorithms, MO shows the best performance in terms of accuracy. This is attributed to the reasonable transition and balance from the exploration phase to the exploitation phase in the MO.

### 3.1.5 Non-parametric statistical analysis

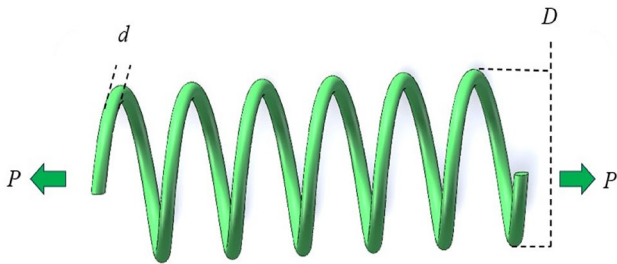
The comparison of the test results between the MO and other optimization methods, as evaluated using the Wilcoxon sign rank tests, is presented in Table 7. It shows that MO's optimization performance is equal to, less than, and better than the comparative approaches, respectively, indicated by the symbols “+”, “-”, and “=”. It can be easily observed that MO is superior to other algorithms in most circumstances. Specifically, out of 161 experimental results, 158 experiments demonstrate the superior performance of MO over other approaches.

## 3.2 CEC 2021 analysis

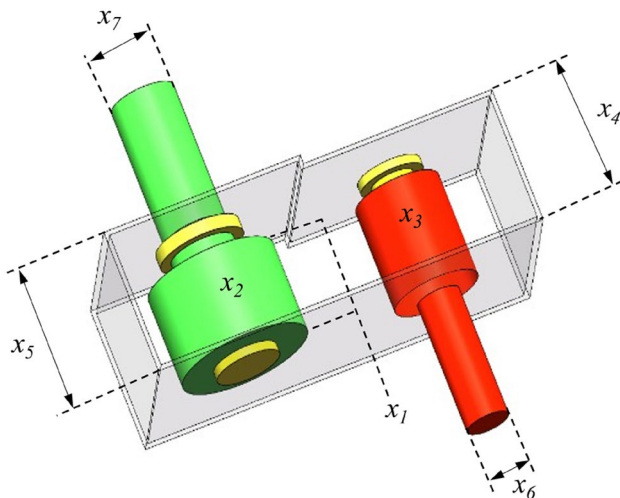
In this section, DMOA [57], GOA [58], MFO [54], MVO [55], NGO [59], SA [60], SO [61], SS [3], and AO [62]

**Table 10** Results of the PVD

| Algorithm | Ts        | Th          | R         | L          | F(x)*       |
|-----------|-----------|-------------|-----------|------------|-------------|
| TTAO [68] | 0.20573   | 3.470489    | 9.036624  | 0.20573    | 5907.851851 |
| SNS [63]  | 0.8125    | 0.4375      | 42.09845  | 176.6366   | 6288.745    |
| HPSO [63] | 0.8125    | 0.4375      | 42.0985   | 176.6365   | 6059.7144   |
| COA [69]  | 0.783,194 | 0.3,874,305 | 40.5794   | 196.3323   | 5893.1336   |
| MSA [69]  | 0.819006  | 0.441004    | 42.43535  | 178.0534   | 5928.544    |
| MBO [69]  | 0.856754  | 0.424026    | 44.38794  | 158.4219   | 6049.427    |
| BKA [70]  | 0.778433  | 0.384690    | 40.319619 | 200        | 5887.364927 |
| ACO [71]  | 0.8125    | 0.4375      | 42.098355 | 176.63772  | 6059.7257   |
| AOA [72]  | 0.8303737 | 0.4162057   | 42.75127  | 169.3454   | 6048.7844   |
| HHO [67]  | 0.817584  | 0.407293    | 42.091746 | 176.719635 | 6000.462590 |
| SMA [73]  | 0.793100  | 0.393200    | 40.671100 | 196.217800 | 5994.185700 |
| MO        | 0.7784    | 0.3848      | 40.3316   | 199.8338   | 5885.7317   |

**Fig. 10** A model of the TSD**Table 11** Results of the TSD

| Algorithm | d         | D         | N          | F(x)*      |
|-----------|-----------|-----------|------------|------------|
| SMA [73]  | 0.049736  | 0.338531  | 11.9671    | 0.012795   |
| BWO [74]  | 0.0517    | 0.3568    | 11.3132    | 0.012703   |
| COA [69]  | 0.0519129 | 0.362122  | 10.9792    | 0.012666   |
| MSA [69]  | 0.04956   | 0.307678  | 14.86782   | 0.013314   |
| MBO [69]  | 0.04956   | 0.313172  | 14.10088   | 0.012935   |
| BKA [70]  | 0.051173  | 0.344426  | 12.047782  | 0.01267027 |
| WOA [52]  | 0.051207  | 0.345215  | 12.004032  | 0.0126763  |
| GSA [19]  | 0.050276  | 0.323680  | 13.525410  | 0.0127022  |
| MO        | 0.0516507 | 0.3557959 | 11.3432253 | 0.0126653  |

**Fig. 11** A model of the SRD

will be tested alongside MO. The data in Table 8 indicates that MO achieves the best results among all methods in the vast majority of experiments. Specifically, the MO obtains optimal results in 6 test functions, namely CF1, CF5, CF6, CF7, CF9, and CF10. Its average fitness values (Avg) are all theoretical optimal values with a standard deviation (Std) of 0, which demonstrates that it not only has extremely high optimization accuracy but also exhibits strong stability in multiple independent runs. In the three functions CF2, CF3, and CF4, the MO ties for first place with the AO algorithm and the NGO algorithm. However, when considering the performance across other functions, the MO has more comprehensive advantages. For example, in CF5 to CF7, the performance of AO and NGO declines significantly (e.g., the average fitness value of AO in CF5 is  $1.28\text{E}+01$ , and that of NGO in CF7 is  $3.85\text{E}-04$ ), while MO still maintains the optimal result of 0, which further highlights its adaptability. In summary, the MO has achieved excellent results of “6 optimal, 3 tied for first, and 1 third” in CEC2021, fully demonstrating its versatility and reliability in handling complex optimization scenarios.

## 4 Practical engineering design problems

In order to comprehensively and thoroughly evaluate the practicality and effectiveness of MO, this work has meticulously selected six typical engineering examples as cases. The comprehensive mathematical formulations pertaining to each of the problems being examined are clearly outlined in the original paper [63].

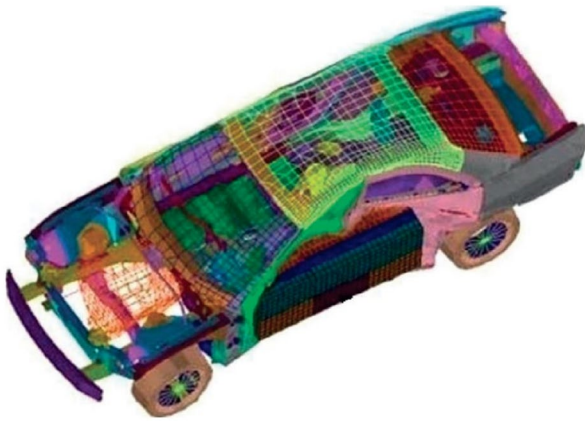
### 4.1 Design of Three-Bar truss (TTD)

In recent years, the TTD is a popular problem, and a lot of energy has been invested in this problem. In Fig. 8, the problem's structural parameters are displayed.

Table 9 compares the results of the MO this problem for with 6 algorithms. In all the competitive results, MO,

**Table 12** Results of the SRD

| Algorithm | $x_1$ | $x_2$ | $x_3$  | $x_4$ | $x_5$ | $x_6$ | $x_7$ | $F(x)^*$ |
|-----------|-------|-------|--------|-------|-------|-------|-------|----------|
| COA [69]  | 3.500 | 0.700 | 17.000 | 7.300 | 7.800 | 3.350 | 5.287 | 2996.348 |
| CPA [69]  | 3.526 | 0.700 | 17.000 | 8.379 | 7.808 | 3.372 | 5.297 | 3035.367 |
| MSA [69]  | 3.506 | 0.700 | 17.000 | 8.309 | 7.808 | 3.358 | 5.295 | 3012.079 |
| MBO [69]  | 3.514 | 0.700 | 17.000 | 7.418 | 7.824 | 3.363 | 5.295 | 3009.238 |
| AMA [46]  | 3.500 | 0.700 | 17.000 | 7.300 | 7.800 | 3.350 | 5.287 | 2996.348 |
| SNS [63]  | 3.502 | 0.700 | 17.000 | 7.300 | 7.715 | 3.350 | 5.287 | 2994.471 |
| SHO[63]   | 3.502 | 0.700 | 17.000 | 7.300 | 7.815 | 3.351 | 5.289 | 2998.551 |
| EAO[75]   | 3.520 | 0.700 | 17.000 | 8.300 | 7.716 | 3.350 | 5.287 | 2994.480 |
| MO        | 3.500 | 0.700 | 17.000 | 7.300 | 7.715 | 3.350 | 5.287 | 2994.471 |

**Fig. 12** A model of the VSID

DEDS, SSA, PSO-DE and HHO won the first place, MBA and CS were the second and third place respectively.

## 4.2 Design of pressure vessel (PVD)

This engineering challenge attempts to achieve four requirements while lowering the cost of manufacturing cylindrical pressure vessels. In Fig. 9, the problem's structural parameters are displayed.

Table 10 compares the results of the MO this problem for with 11 algorithms, including TTAO, SNS, HPSO, COA, MSA, MBO, BKA, ACO, AOA, HHO and SMA. Obviously, the competitive result of the MO gets the first place among all the algorithms. The BKA algorithm also performed very well in this case, ranking second.

## 4.3 Design of tension/compression spring (TSD)

The aim of this engineering challenge is to successfully meet three crucial and potentially conflicting requirements, all while striving to significantly reduce the weight of the coil. Figure 10 displays the problem's structural parameters.

The running results of MO with other methods in this case are shown in Table 11. We can find that the MO run results perform the best among all the data. Specifically,

COA, WOA, GSA and SMA perform equally well and are ranked next to MO, while MBO and MSA run results are relatively poor.

## 4.4 Design of speed reducer (SRD)

This problem is categorized as a discrete optimization task, aimed at satisfying 11 distinct criteria while achieving a reduction in the weight of the reducer device. Figure 11 visually illustrates the structural parameters that constitute this problem.

Table 12 compares the results of the MO this problem for with 9 algorithms, including COA, CPA, MSA, MBO, AMA, SNS, SHO and EAO. In all experimental results, the MO and SNS achieved the optimal values, the experimental result of the EAO ranked second, while the performance of CPA was unsatisfactory.

## 4.5 Design of vehicle side impact

The Vehicle Side Impact Design (VSID) aims to minimize the weight of the door to the greatest extent using nine influence parameters [76]. Figure 12 shows a model for VSID.

Table 13 lists the results of the VISD. The optimization results provided by MO are the most feasible among the seven methods. Specifically, MO can achieve the best results in VISD, while WSO gets the worst results. This indicates that MO has strong competitiveness when dealing with complex problems.

## 4.6 Design of piston lever (PLD)

The prime objective of this problem lies in decreasing the fuel consumption throughout the movement, particularly when the object is elevated from an initial  $0^\circ$  angle to  $45^\circ$  [78]. Figure 13 shows a model for PLD.

Table 14 presents the results of the six approaches in solving the PLD problem. Among them, the experimental results of CS, SNS, CSGO1, HSO-GA and SGA are all from the reference [78]. The data in Table 14 illustrates that MO can offer the best solution in terms of solving the PLD



optimization problems. After a large number of experimental results comparisons, the feasibility, scientific and superiority of MO are proved.

Although the MO demonstrates excellent comprehensive performance, it has certain limitations. For instance, the experimental results of the MO on F5, F6 and F13 do not stand out among numerous optimization algorithms, which may be due to the insufficient exploration of the behavioral patterns of musk ox populations. In future research, the MO can be combined with other social behaviors of musk ox populations and other well-known strategies to further enhance its performance. Furthermore, the MO can be expanded to address challenges related to multidisciplinary design optimization and multi-objective optimization.

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**Data availability** Data is provided within the manuscript or supplementary information files.

## Declarations

**Conflict of interest** The authors declare that there is no Conflict of interest. Non-financial Conflict of interest.

**Ethical approval** In the research covered by this article, none of the authors have conducted any experiments involving human participants or animals.

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